

**FIRST PERSON  
TO WIN  
TWO  
NOBEL  
PRIZES IN  
CHEMISTRY**



**SEQUENCED  
48,502  
BASE  
PAIRS OF  
DNA BY  
1982**

was the first protein to be sequenced, and there followed a surge in protein research, which later enabled the development of synthetic insulin for treating diabetes.

**Genetic research**

Following the award of the Nobel Prize in Chemistry in 1958 for his work on insulin, Sanger had the freedom to pursue his most ambitious research goals and, in 1962, he moved to the Laboratory of Molecular Biology in Cambridge. Having caught what he called “the sequencing bug,” he now turned his attention to exploring the structure of the body’s genetic nucleic acids, RNA and DNA.

Sanger’s Cambridge colleagues Francis Crick and James Watson had discovered the double-helix structure of DNA—the master chemical containing instructions for cell growth and function, as well as inheritable genetic characteristics—in 1953. Sanger aimed to untangle this structure to reveal the bases in specific strands of DNA, each of which carried distinct genetic code. Beginning with RNA, a smaller and less complex single-stranded nucleic acid that relays instructions and creates proteins

**“I had 20 years  
when I could  
just do what  
I wanted.”**

**Frederick Sanger, 2001**

according to the information encoded in DNA, Sanger developed a sequencing method using radioactive isotopes to “label” individual fragments and build up the sequence by looking at the areas of overlap between them. By 1967, he had a complete sequence of RNA from *E. coli* bacteria.

**Unlocking the code**

In the early 1970s, Sanger moved on to DNA, which was harder to sequence due to its large size and double-helix structure. The sequencing breakthrough came in 1977, when his team pioneered a method of isolating DNA fragments via dideoxy chain-termination—using a molecular inhibitor to prevent DNA

